NPLify · P0 Technical Baseline

Test Strategy & Coverage

New XP Technologies Limited
21 October 2026 · Confidential · new document, added for the design review (Pete, points 6a to 6d; Andrew, items 2.5 and 9.4)

1 · What this document is for

The pack already contains the yardsticks: the acceptance vectors of the Calculation Specification, the worked transactions of the Ledger Posting Design (61 numbered, T1 to T61; 67 fixtures once the variants T3a, T22a, T24a, T41a, T41b, T41c and T52a are counted, T41 existing only as its three variants — the Reading Companion's "67 worked transactions" is the same set; Appendix A is the one manifest), the state machines of the Lifecycle & Approvals document, the field groups of the Roles & Visibility Matrix, the constraints of the ERD's Appendix C and the exit criteria of the Cutover Approach. It did not say in one place which tests exist, what each proves, where it runs and what passing means. This document does. It is the testing framework Pete asked for: full scope, the purpose and expected result of every test, regression so the system stays robust as the project evolves, and a map from the Workflow Tree to the tests, so that well-tested and under-tested areas are visible.

2 · Test levels and where each runs

Level	What it proves	Where it runs	Who writes it	When
unit	One function gives the expected figure: pricing, rounding, breach check, tier split	in the build, on every change	Engineering, from the vectors	from M0
fixture	A posted event produces exactly the lines of a worked transaction, and the identities hold afterwards	in the build, against a fresh database	Engineering, from the 67 fixtures of Appendix A	from M2
database	A write that breaks a constraint is refused by the database itself, not only by the application	in the build, against the migrated schema	Engineering, from Appendix C	from M2
integration	A sequence of actions through the API follows the state machine, the approvals and the visibility rules	in the build and on staging	Engineering, from the Lifecycle and Roles documents	M3 onward
release	A release can be applied and rolled back; migrations run against last week's snapshot with the ledger checks green	staging, before every release	Engineering	every release
UAT	NPL staff run real procedures with real evidence and get the result the SOP says	staging	NPL Operations and Finance, with New XP	M10
parallel run	The app reproduces the sheets, day after day, to the cent	staging	NPL Finance	M10, three to four weeks
acceptance	The go-live criteria 1 to 13 of the Cutover Approach	staging, then production on the switch day	Engineering with NPL sign-off	M10, M11

3 · The acceptance gate per phase, and the regression rule

Every feature moves through the proposal's pipeline (mockup, build, QA, staging, UAT, production). Each phase closes with a demonstration against its gate; the gate is the set of tests below, all green. The regression rule: every release, in every phase and under the support bundle after go-live, runs the full vector, fixture, state and visibility suites, and a failing exact test blocks the release. A test is never removed because it fails; it is fixed or its expected result is changed with a recorded reason and NPL's agreement where the figure is NPL's.

Phase gate	Must be green
Phase 1 (M0 to M5): the calculating core	V-L1…L14, V-B1…B23, V-RB1…4, V-ED-A…D, V-TIER1…3, F-T1…T61 (67 fixtures, Appendix A) as far as the patterns are built, F-ID1…12, F-REPLAY, F-REV, D-CONS1…28, D-POLY1…7, S-DEAL, S-SETT, S-RATE, S-RR, C-APPR, C-JOBS, C-ATOM; the P-tests of the class 4 and 5 procedures at their gate (6a); the end-to-end demonstration reproduces the reference workbook figures
Phase 2 (M6, M7): hardening	everything above plus R-VIS1…n, R-ACCESS, C-LOCK, S-EMERG, C-INT1…3 on staging; the P-tests of the class 6 and 7 procedures and of 1.4 (6a); the role-gated, reconciled platform with side-by-side comparison reports
Phase 3 (M8 to M11): completion and go-live	everything above plus O-EVID1…3, O-C6A, O-SLA, G-SEC, G-CONC, G-OUT, G-DUP, G-EXT, G-REST, G-OPEN, G-FILTER, G-RECON, G-VULN, D-MIG; the go/no-go on Finance's discrepancy log
After go-live (support bundle)	the full suite on every release; G-REST rehearsed once a year; the coverage sheet reviewed at every quarterly access review

4 · Identifiers and result classes

Every test has an identifier that never changes and is never reused (the same rule as the T, L, B, W, F and C identifiers of the pack; the Reading Companion's register lists them). The first letter says the family: V vectors and calculation, F fixtures and identities, S state machines, C concurrency and controls, D database, R roles and visibility, O operating procedures and UAT, G go-live acceptance', mk(', P the per-procedure tests of Section 6a (P-<procedure>-pos, -den, -rtr)'), '. A purpose line says what the test proves. A result class says what passing means, so that a reviewer can tell a test with one right answer from a test whose result must only be reasonable:

Result class	Passing means	Examples
exact	The result equals the expected value to the cent (eight decimals for BTC amounts, the unit where NPL's sheet rounds), or the expected state or refusal occurs	V-L1, F-T2, S-DEAL, D-CONS
tolerance	The result is within the rounding of the pair (a dropped fraction or a round-up with its owner)	cash rounding cases; margin-of-the-day reconciliation
range	The result falls inside a configured band, because the right answer depends on data or judgement	G-RECON checks with a tolerance; G-VULN severity; aging thresholds
invariant	A property that must always be true holds after every step and every fixture	F-ID1…12, F-REPLAY, C-APPR, R-VIS, G-DUP

5 · Test inventory

One row per test or test family; the family's members are numbered in the test suite itself (vectors-v5.9.csv for V, the fixtures for F). The source column names where the expected result is written down in the pack; the phase column when the test first has to be green.

Id	Purpose	Result class	Level	Source in the pack	Phase
V-L1 … V-L14	The pricing function reproduces each acceptance vector A: sender rate, parts, client net, delivered, earnings, payable, residual (L14, round 2: the opening volume in the counter)	exact (to the cent; BTC to 8 decimals)	unit	Calculation Specification §18, 18a; vectors-v5.9.csv	M1, M4

V-B1 ... V-B23	The pricing function reproduces NPL's own figures: the sheet lines B1 to B18 (B2 and B18 against the corrected sheet of 21 October) and the five figures B19 to B23 NPL supplied on 21 October for the 18a cases and the same-currency deal	exact (to the unit where the sheet rounds)	unit	Calculation Specification §19, 19a, 19b	M1, M4
V-RB1 ... V-RB3	Breach rules 2a: thin history → fallback threshold; zero or negative baseline → no percentage formula; absolute tolerance either way → flagged, not frozen (flagged: unusable until Finance confirms; the pair not frozen)	exact (breach / flag / neither)	unit	Calculation Specification 2a; THRESHOLD kinds	M1
V-RB4	A quote and a conversion attempted against a flagged rate version are refused and the refusal is in AUDIT_LOG (quote_refused, conversion_refused); after Finance confirms the version with a reason, the same quote prices on it	exact (refused / priced; audit row present)	integration	Calculation Specification 2a; Lifecycle Figure 3	M1
V-ED-A ... V-ED-D	Effective-date scenarios A to D: a fee change on 1 November applies only to deals priced from then; fees, tiers, rounding and partner settings	exact	integration	Configuration Schema 1a; Lifecycle 2a	M4
V-TIER1, V-TIER2	Tier exactly reached and tier crossed on GDC (L12, L13); a void lowers the counter; two simultaneous quotes read the counter in order	exact	integration	Calculation Specification 5a; ERD Appendix D	M0, M3
V-TIER3	Opening tier volume (L14): the approved opening line seeds the counter and the deal of the opening day splits at the threshold; an in-flight deal keeps its pre-opening priced_at and is not counted twice; a void of such a deal lowers the counter; the opening month's turnover report shows the opening line once and the next month not at all	exact	integration	Calculation Specification 5a; Cutover §4; OPENING_ENTRY month_volume	M3
F-T1 ... F-T61 (67 fixtures)	Each worked transaction posts exactly the lines, currencies, ownership tags and components of the Ledger Posting Design, and the balances after it; one fixture per row of the manifest in Appendix A, variants included (F-T3a, F-T22a, F-T24a, F-T41a/b/c, F-T52a); the build refuses a manifest row without a fixture and a fixture without a manifest row	exact	fixture	Ledger Posting Design T1 to T61; Reading Companion register; Appendix A; fixtures-manifest-v6.0.csv	M2
F-ID1 ... F-ID12	The identities of Section 20 hold after every fixture: client assets = liabilities per currency; PAYABLE = Σ GROUP_ENTITLEMENT; partner balance = client + pool; rebates and shares agree to accruals; losses close; own wallets hold client money in four cases only; every released line is accounted for	invariant	fixture	Ledger Posting Design §20	M2
F-REPLAY	A retried event with the same idempotency key posts nothing (rule 1a), for every source type of §22	invariant	integration	Ledger Posting Design 1a; Architecture §5	M2
F-REV	A correction is a reversal naming what it reverses plus a re-posting, dated today with corrects_period (rule 11a); a reported month never changes	invariant	integration	Ledger Posting Design 11a, Pattern E	M2
S-DEAL	Every transition of Figure 1 with its guard and its approver; every refused transition (wrong state, wrong role, same person) refused	exact	integration	Lifecycle §2, 2a, 2b, 7, 7a	M3 to M5

S-SETT	Settlement run and lines: approve, release, confirm full and short, return; "sent, not yet confirmed" shown and aged; after release nothing is cancelled	exact	integration	Lifecycle §3, Figure 2; Ledger rule 13, 13a	M5
S-RATE	Rate version states: breach freezes the pair; Finance approval with a reason; expiry; a flagged version is unusable until Finance confirms it (V-RB4)	exact	integration	Lifecycle Figure 3	M1
S-RR	Reroute before and after conversion: custody bounded, forward, dues recovered, Management ratification for post_conversion	exact	integration	Lifecycle Figure 4, 2b; Pattern O	M5
S-EMERG	Emergency suspend / freeze / cutoff override takes effect at once, blocks what it must, is lifted if not ratified within a business day	exact	integration	Lifecycle §5 addition; ERD EMERGENCY_ACTION	M6
C-APPR	Approval valid only for the record version seen; two approvers at once: one wins; approver's person_ref never the initiator's	invariant	integration	Lifecycle 6a; Roles 5a	M3 to M6
C-JOBS	A job started twice runs once; catch-up after a restart; bounded retries; the job-health line	invariant	integration	Architecture §4 addition	M1 onward
C-ATOM	Atomicity of posting (Ledger rule 1b): a failure is injected at every step of every posting path of Section 22 — before the source state change, between it and the transaction row, between the row and its postings, before and after the partner call — and after each: no source record is in a state its posting does not match, no LEDGER_TRANSACTION is duplicated (the unique index on the idempotency key and on source event refuses it), a retry with the same key on a new request returns the first result, and the only state a failure after the partner acted leaves is an instruction with no posting, which the exception list shows and Pattern P case 1 repairs once	invariant	integration	Ledger Posting Design 1a, 1b, 19b; ERD Appendix C	M2
C-LOCK	A record lock is released on save or cancel, expires after lock_idle_minutes, can be cleared by Management, and every step is in AUDIT_LOG	exact	integration	Lifecycle 6a; ERD RECORD_LOCK	M6
C-INT1 … C-INT3	Interruption cases of Pattern P: partner executed and the application failed; confirmation never arrives; explorer unavailable. No duplicate, no posting on a guess. G-EXT is the acceptance-level test of the same situation seen from the partner side	invariant	integration / staging	Ledger Posting Design 19b	M7
D-CONS1 … D-CONS32	Each rule of ERD Appendix C: a write that breaks it is refused by the mechanism named (key, CHECK, trigger or validation)	exact (refused)	database	ERD Appendix C	M2
D-CLONE	A project cloned from an Evo-like source for a partner on another rail: the clone leaves the reconfirm list empty, keeps the source on every copied row, and activation is refused with field-level errors (superseded fee version, missing settlement rail, unconfirmed fee percentage and partner fee); after the fixes it validates and Management activates it; a partner cutoff is never copied	exact (refused with the named errors; then activated)	integration	Configuration Schema 1a, §6 rule 15	M0

R-SCREEN	Wallet screening: a report clear under the Strict rulebook (score in the clear band, no category at threshold, no terrorism / TF trace) lets Operations accept; any other report goes to Management's second look and no Finance user can change a reject; a partner's hold, freeze or decline is recorded as its event and cannot be lifted at NPL by any role through screen, API, export re-import or a replayed approval request (Roles rule 11); a re-screen is a new row with evidence, the old row untouched	exact (who may decide; refused for Finance)	integration	Configuration Schema §4 WALLET_SCREENING, §6 rule 10; Roles §5	M4
D-POLY1 … D-POLY7	The generic trigger refuses a type-plus-id reference whose id does not exist, for each of the seven references	exact (refused)	database	ERD Appendix C, option b	M2
D-MIG	Every migration runs against last week's restored snapshot with the ledger checks green; ledger tables additive only	invariant	release	Architecture §6 schema change policy	every release
R-VIS1 … R-VISn	Visibility by field group, per table and per role, on every read, list, export, document, notification, error message and the client journal: Operations never receives REF, ECON or LEDGER fields; includes the two fallback workbooks and their share lists, generated extracts, monitoring alarms, chat messages and incidents on the delivery board (Roles rule 12)	invariant	integration	Roles §2, §5 rule 2, 3, 12; Architecture §5	M6
R-ACCESS	The quarterly access review report lists every account with person_ref, last sign-in and open approvals; a dormant account can be disabled and nothing is deleted	exact	integration	Operating Procedures A.10; Roles 5a	M6
R-BANK	Bank account details read in full and logged: a user of any role opens the bank-details card of an account, endpoint, rail or destination approval, or exports it; every field of the current BANK_ACCOUNT_DETAILS version is shown unmasked; one AUDIT_LOG row (record_type BANK_ACCOUNT_DETAILS, field read, actor, time) exists per full read, card, export or API alike; a superseded version is shown only from the history and its read is logged the same way	exact (fields shown; audit row present)	integration	Roles §5 rule 10; ERD Appendix C	M6
O-EVID1 … O-EVID3	Evidence checklist: a collection cannot be verified without the required evidence or an authorised exception; hash missing; cash short; partner says paid while the receiver disputes	exact	UAT	Operating Procedures A.11, D.2, D.3	M10
O-C6A	Rejected payout: each of the four outcomes follows its steps and approvers and posts as Pattern F	exact	UAT	Operating Procedures C6a	M10
O-SLA	A pending item past its approval_time_limit appears on the escalation role's list and raises an alert; cutoff warnings at the configured minutes	exact	integration	Lifecycle 8a; THRESHOLD kinds	M8
G-SEC	Go-live criterion 9: an Operations user tries to see a margin or approve a settlement through screens, API, export, notification and journal: denied, nothing changed	invariant	acceptance	Cutover Approach exit criteria 9 to 13	M10

G-CONC	Criterion 10: two users edit the same deal; two Finance users approve the same settlement; a job started twice	invariant	acceptance	Cutover Approach	M10
G-OUT	Criterion 11: the application is stopped mid-collection and mid-conversion; on restart nothing lost, nothing half-posted	invariant	acceptance	Cutover Approach	M10
G-DUP	Criterion 12: the same settlement instruction submitted twice; the same collection recorded twice — the duplicate arises on NPL's side; the partner-side case is G-EXT	invariant	acceptance	Cutover Approach	M10
G-EXT	Unknown external outcome: a settlement or conversion instruction is sent and the partner's answer never arrives, or arrives after the application has timed out (the partner accepted it; the partner never received it; the partner answered and the answer was lost). Expected: the line goes to the state "sent, outcome unknown" and is held; no second instruction is emitted by a retry, a job or a user; Operations raises a partner reference inquiry and records the partner's answer with evidence; only when the outcome is established does anything move — executed: confirm against the partner's reference and post nothing new; not received: Finance authorises one retry under a new partner reference, recorded as a retry of the held line. Simulated on staging with a partner stub that accepts and does not answer, and one that never receives: zero duplicate instructions in the stub's log, and the exception reconcilable from the partner's statement alone (Pattern Q)	invariant	acceptance	Ledger Posting Design 19b (Pattern P, C-INT2); Lifecycle & Approvals 13a and Figure 2; Cutover Approach criterion 12	M10
G-REST	Criterion 13: restore to a point in time and recover under the runbook 8a; two sign-offs; no missing or duplicate movement	invariant	acceptance	Cutover Approach 8a	M10, rehearsed yearly
G-OPEN	Opening entries: every Section 5 check agrees exactly with the sheets as of the opening date; in-flight deals continue in the app; one deal imported in each state of the per-state opening mapping (Cutover §4), continued to its close: every balance changes exactly once, nothing already in the opening entries posts again; a suspense line above the materiality figure fails the go/no-go on its own	exact	acceptance	Cutover Approach §4 (per-state mapping, step 3a); Pattern R	M11
G-FILTER	The clean-target extract: a dummy project with a collection, a conversion, a payout, a reversal and evidence on staging; production built by the extract of Cutover §7 has no row linked to it in any table (checked through the foreign keys), the real projects' per-table counts and balances equal staging's, every evidence reference and ledger hash resolves, and nothing was deleted on staging	exact	acceptance	Cutover Approach §3, §7	M11
G-RECON	Ten consecutive business days of the Section 5 checks inside tolerance; a reconciliation record per partner × currency × day	range (tolerance per check)	parallel run	Cutover Approach §5, §6; Pattern Q	M10
G-VULN	Basic vulnerability assessment of staging; critical and high findings fixed before production	range (severity)	security	Architecture 6a	before go-live

6 · Coverage sheet: the Workflow Tree against the tests

One row per sub-class of the Workflow Tree (27 sub-classes, 92 procedures). For each: where the procedures are written up, which client scenarios of the Scenario Catalogue use them, which ledger pattern books them, which tests exercise them and how many. A count of zero would be an open area; today none is zero, but the two lowest rows (6.2 payout methods, 7.5 ledger corrections) are where Phase 3 adds UAT cases first. The sheet is kept as a file next to the test suite (coverage.csv), with one row per procedure (the 92 leaves) filled in as the build phases land, and is shown at every phase review; the sub-class view here is its summary at the start of the build. Note: coverage.csv is a build-phase deliverable (Phase 1 onwards, maintained through Phase 3), not part of the P0 review cycle; its test column can only be filled once the test suite exists. From this edition the sheet also records, per procedure, the tests executed at the last phase review and their results, not only the families planned: a count here is a count of tests that ran and passed.

Workflow Tree sub-class	Written up	Client scenarios (Catalogue)	Ledger pattern	Tests	Count
1.1 Start a new project	OP Part C; CS 1a	all	—	V-ED, D-CONS (project state), D-CLONE	3
1.2 Change configuration	CS §1 change control, 1a	all	—	V-ED-A…D, D-CONS (write-once)	5
1.3 Parties, endpoints and destinations	CS §4, §5; OP C2, C3, C7	1.1–1.6, 2.1–2.5, 4.1–4.4	—	D-CONS (accounts, endpoints, single-use), R-VIS, R-BANK	5
1.4 Users, roles and access	Roles §5, 5a; OP A.10	all	—	C-APPR, R-VIS, R-ACCESS, P-1.4.1, 1.4.2 (pos, den)	7
2.1 Partner rate versions	Calc §2, 2a; LC Figure 3	all converting scenarios	—	S-RATE, V-RB1…4	5
2.2 Market reference	Arch §4 rate capture; CS RATE_SOURCE	all	—	C-JOBS, V-RB	2
2.3 Bank cutoff day (C9)	OP C9; CS PARTNER; LC §2	2.6, 4.1, 4.3, 6.2	K (T37)	S-DEAL (cutoff), F-T37	2
3.1 Price and send a quote	LC §2; Calc §3–§7; X1	2.1–2.5, 4.x	A	V-L1…L9, V-B1…B23, S-DEAL	28
3.2 Validity, grace and late decisions	LC §2, OP A.2, D.1	2.4	K (T38)	S-DEAL, F-T38	2
3.3 Entry modes without a quote	LC §2; SC W2	2.4, 6.1	A	S-DEAL (collect-first), V-ED-D	2
3.4 Fee mechanics decided at quote	Calc §4, 4a, §5, 5a, §6, §11	1.4, 1.5, 4.3, 5.1	J	V-TIER1…3, V-L6, V-L12, V-L13, V-L14, F-T30…T35	12
4.1 Collection methods	OP C1, C2, C4, C7; X2	1.x, 2.x, 3.2, 4.2, 5.x	A, B, D	O-EVID1…3, F-T1, T4, T5, T15…T17, P-4.1.1…4.1.5 (pos, den, rtr), R-SCREEN	25
4.2 Verify the collection	OP A.3, A.11; LC §5	all	A, I	O-EVID, F-T28, T29, D-CONS (endpoint), P-4.2.1…4.2.4 (pos, den; rtr on 4.2.1, 4.2.2)	15
5.1 Conversion on a deal leg	LC §2, §5; Calc §8, §12; X3	all converting	A, K	F-T2, T3, T6, T11, T14, T36, S-DEAL, P-5.1.1…5.1.6 (pos, den; rtr on 5.1.1, 5.1.2, 5.1.5, 5.1.6), G-EXT	24
5.2 Conversions not on a deal	LC §4; Calc §16; X6	6.2	H	F-T26, T27, S-DEAL (balance conversion), P-5.2.1 (pos, den, rtr), G-EXT	7
6.1 Prepare, approve and release	LC §3, Figure 2; OP A.5; X4	all paying out	C	F-T7…T10, S-SETT, O-SLA, G-DUP, P-6.1.1…6.1.4 (pos, den, rtr), G-EXT	20
6.2 Payout methods	OP C1, C3, C5, C8, C10	1.3, 1.6, 3.3, 4.4, 5.x, 6.3	C	O-EVID, F-T23, P-6.2.1…6.2.6 (pos, den, rtr), G-EXT	21

6.3 Bank fees and returns	OP C6, C6a, D.2, D.3; LC §3	2.7, 4.1	C, F	F-T9, T10, T22…T25, O-C6A, P-6.3.1…6.3.6 (pos, den; rtr on 6.3.1, 6.3.3, 6.3.4, 6.3.5)	23
7.1 Reroute and custody	LC Figure 4, 2b; OP A.8; X5	1.x (Ali unavailable), 2.5	D, O	F-T12…T17, T56…T58, S-RR, C-INT1, P-7.1.1…7.1.4 (pos, den; rtr on 7.1.1…7.1.3)	22
7.2 Voids and unwinds	LC §5; OP D.4; Ledger E	—	E	F-T18…T21, S-DEAL (voids), P-7.2.1…7.2.3 (pos, den, rtr)	14
7.3 Loss events and recoveries	LC Figure 6; OP C7, D.5	4.2	L	F-T39…T43, P-7.3.1, 7.3.2 (pos, den, rtr)	13
7.4 Earnings resting at partners	LC Figure 7; Ledger N; X6	1.x, 2.x	N	F-T52…T55, S-RR (dues), P-7.4.1, 7.4.2 (pos, den, rtr)	11
7.5 Ledger corrections	Ledger E addition, rule 11a	—	E	F-REV, F-T19, P-7.5.1 (pos, den, rtr)	5
8.1 Daily	OP A.6, A.12; Cutover §5	all	P, Q	G-RECON, C-INT1…3, C-JOBS, G-EXT	7
8.2 Month end	OP A.7; Ledger §20, Pattern Q; X7	1.x, 3.3	J, M, Q	F-T31, T34, T35, T44…T51, T59, F-ID1…12	25
8.3 Reports the ledger answers	Ledger §21; Roles	all	—	R-VIS, F-ID	2
8.4 Cutover: parallel run, opening entries and switch	Cutover §3, §4, §6, §7, 8a; Pattern R	all	R	G-OPEN, G-FILTER, G-REST, G-RECON, F-T60, T61	6

6a · Per-procedure traceability: the money-moving, permission and exception procedures

A sub-class count above does not prove each of its procedures. For the procedures that move money, grant a right or handle an exception (classes 4 to 7 of the Workflow Tree, the two user-and-rights procedures of 1.4, and the two rights-bearing quote procedures 3.2.5 and 3.4.2) the test suite carries one explicit test per case and per procedure, named P-<procedure>-pos, -den and -rtr. The positive case (-pos) is the same for every row: the procedure completes with the initiator and approver the tree names, and the record, approval and posting it names exist exactly once. The gate is the milestone at which all of the procedure's cases must have run and passed; a procedure whose cases have not run does not pass its gate, whatever its sub-class count says. The denied and retry columns use codes: S — refused in the wrong state; R — refused for a role without the right (ROLE_RIGHT); P — refused when the approver is the person who initiated; M — refused until the Management approval exists (Management "always"; M(cond) where the tree makes it conditional); D — the same submission twice with the same idempotency key acts once and the second returns the first result (rule 1a); X — the partner's outcome is unknown: the line is held and no second instruction leaves (G-EXT).

Procedure (Workflow Tree)	Positive case	Denied cases	Retry	Gate
1.4.1 create, change or remove a user or role	P-1.4.1-pos	P-1.4.1-den: S R P M	—	M6
1.4.2 set approval rights per controlled action	P-1.4.2-pos	P-1.4.2-den: S R P M	—	M6
3.2.5 honour an expired quote's rate on the new deal	P-3.2.5-pos	P-3.2.5-den: S R P	—	M3
3.4.2 raise the fee above the markup cap	P-3.4.2-pos	P-3.4.2-den: S R P M	—	M3
4.1.1 C2 · USDT collection with wallet screening	P-4.1.1-pos	P-4.1.1-den: S R P	P-4.1.1-rtr: D	M4
4.1.2 C4 · bank transfer in	P-4.1.2-pos	P-4.1.2-den: S R P	P-4.1.2-rtr: D	M4
4.1.3 C1 · cash by token (collection side)	P-4.1.3-pos	P-4.1.3-den: S R P	P-4.1.3-rtr: D	M4
4.1.4 C7 · one-time local deposit account	P-4.1.4-pos	P-4.1.4-den: S R P	P-4.1.4-rtr: D	M4
4.1.5 bridge collection on NPL's own wallet	P-4.1.5-pos	P-4.1.5-den: S R P M	P-4.1.5-rtr: D	M4

4.2.1 verify a collection part	P-4.2.1-pos	P-4.2.1-den: S R P	P-4.2.1-rtr: D	M4
4.2.2 overpayment → sender credit	P-4.2.2-pos	P-4.2.2-den: S R P	P-4.2.2-rtr: D	M4
4.2.3 underpayment → wait or void	P-4.2.3-pos	P-4.2.3-den: S R P	—	M4
4.2.4 instructions issued: accepted → collecting	P-4.2.4-pos	P-4.2.4-den: S R P	—	M4
5.1.1 instruct a conversion	P-5.1.1-pos	P-5.1.1-den: S R P	P-5.1.1-rtr: D X	M4
5.1.2 partner books; margin recognised	P-5.1.2-pos	P-5.1.2-den: S R P	P-5.1.2-rtr: D	M4
5.1.3 keep the deal's margin unconverted	P-5.1.3-pos	P-5.1.3-den: S R P	—	M4
5.1.4 resolve a conversion variance exception	P-5.1.4-pos	P-5.1.4-den: S R P M(cond)	—	M4
5.1.5 two-leg route: hop to the second partner	P-5.1.5-pos	P-5.1.5-den: S R P	P-5.1.5-rtr: D X	M4
5.1.6 own-desk conversion	P-5.1.6-pos	P-5.1.6-den: S R P	P-5.1.6-rtr: D	M4
5.2.1 balance conversion	P-5.2.1-pos	P-5.2.1-den: S R P M	P-5.2.1-rtr: D X	M4
6.1.1 prepare and release a settlement	P-6.1.1-pos	P-6.1.1-den: S R P M(cond)	P-6.1.1-rtr: D X	M5
6.1.2 receiver confirms: full, or short → shortfall	P-6.1.2-pos	P-6.1.2-den: S R P	P-6.1.2-rtr: D	M5
6.1.3 top up a shortfall from the pool	P-6.1.3-pos	P-6.1.3-den: S R P M	P-6.1.3-rtr: D	M5
6.1.4 pay a receiver from another group's leg (offset)	P-6.1.4-pos	P-6.1.4-den: S R P M	P-6.1.4-rtr: D	M5
6.2.1 C3 · receiver wallet screened, then USDT payout	P-6.2.1-pos	P-6.2.1-den: S R P	P-6.2.1-rtr: D X	M5
6.2.2 C5 · bank transfer out	P-6.2.2-pos	P-6.2.2-den: S R P M(cond)	P-6.2.2-rtr: D X	M5
6.2.3 C10 · internal transfer inside the partner's platform	P-6.2.3-pos	P-6.2.3-den: S R P	P-6.2.3-rtr: D X	M5
6.2.4 C8 · local payout for a small amount	P-6.2.4-pos	P-6.2.4-den: S R P	P-6.2.4-rtr: D X	M5
6.2.5 C1 · cash by token (payout side)	P-6.2.5-pos	P-6.2.5-den: S R P	P-6.2.5-rtr: D X	M5
6.2.6 NPL-GR vendor payment discharging a vendor invoice	P-6.2.6-pos	P-6.2.6-den: S R P	P-6.2.6-rtr: D X	M5
6.3.1 bank fee event: waived, carried forward or absorbed	P-6.3.1-pos	P-6.3.1-den: S R P	P-6.3.1-rtr: D	M5
6.3.2 override the bank-fee treatment on one settlement	P-6.3.2-pos	P-6.3.2-den: S R P	—	M5
6.3.3 cover a bank fee for the client	P-6.3.3-pos	P-6.3.3-den: S R P M	P-6.3.3-rtr: D	M5
6.3.4 C6 · returned transfer (bank rejects the payout)	P-6.3.4-pos	P-6.3.4-den: S R P M(cond)	P-6.3.4-rtr: D	M5
6.3.5 open a return leg to the sender	P-6.3.5-pos	P-6.3.5-den: S R P M(cond)	P-6.3.5-rtr: D X	M5
6.3.6 override the network-fee policy on a refund	P-6.3.6-pos	P-6.3.6-den: S R P	—	M5
7.1.1 request a reroute	P-7.1.1-pos	P-7.1.1-den: S R P M(cond)	P-7.1.1-rtr: D	M5
7.1.2 outbound transfer from NPL's own wallet	P-7.1.2-pos	P-7.1.2-den: S R P M	P-7.1.2-rtr: D X	M5
7.1.3 recover dues on a reroute	P-7.1.3-pos	P-7.1.3-den: S R P M	P-7.1.3-rtr: D	M5
7.1.4 custody aging and the daily exception list	P-7.1.4-pos	P-7.1.4-den: S R P	—	M5
7.2.1 void a deal before money moved	P-7.2.1-pos	P-7.2.1-den: S R P	P-7.2.1-rtr: D	M5
7.2.2 void a deal after collection	P-7.2.2-pos	P-7.2.2-den: S R P M	P-7.2.2-rtr: D	M5

7.2.3 unwind a deal after a loss	P-7.2.3-pos	P-7.2.3-den: S R P M	P-7.2.3-rtr: D	M5
7.3.1 record a loss event and approve its split	P-7.3.1-pos	P-7.3.1-den: S R P M	P-7.3.1-rtr: D	M6
7.3.2 record a loss recovery received	P-7.3.2-pos	P-7.3.2-den: S R P	P-7.3.2-rtr: D	M6
7.4.1 earnings offset between partners	P-7.4.1-pos	P-7.4.1-den: S R P M	P-7.4.1-rtr: D	M6
7.4.2 earnings receivable recovered on the partner's fee cycle	P-7.4.2-pos	P-7.4.2-den: S R P	P-7.4.2-rtr: D	M6
7.5.1 ledger adjustment (reversal and re-posting)	P-7.5.1-pos	P-7.5.1-den: S R P M	P-7.5.1-rtr: D	M6

7 · The three files that travel with this document

File	Content	Owner
vectors-v5.9.csv	One row per acceptance vector L1 to L14 and B1 to B23: every input (amount, rates, fee parts, mode, rounding) and every expected figure, as printed in the Calculation Specification; the automated tests read this file, so a change to a vector is a change to this file with a recorded reason	New XP; NPL Finance approves changes to B rows
coverage.csv (build phase)	One row per procedure of the Workflow Tree (92 leaves) with the columns of Section 6 and, per procedure, the P-tests of 6a with the result of their last run; counts are computed from the test suite at each phase review. Created in Phase 1 (M0) from the Workflow Tree, filled as the milestones land; not a P0 deliverable	New XP, from Phase 1
fixtures-manifest-v6.0.csv	One row per fixture of Appendix A (67 rows): fixture id, worked transaction, pattern, source type, number of posting lines, accounts touched, the identities checked after it, the page of the Ledger Posting Design it reproduces, build status. Generated from the Ledger Posting Design, so it cannot drift from it; the build checks that every row has a fixture and every fixture a row (no orphan, no gap)	New XP; regenerated with every edition of the Ledger Posting Design

Identifiers in this document are frozen from this edition. New tests take the next free number in their family; a withdrawn test keeps its number with the note "withdrawn". Added in this edition: the P family (6a), G-EXT, and the fixture ids of the variants (F-T3a, F-T22a, F-T24a, F-T41a, F-T41b, F-T41c, F-T52a).

Appendix A · Fixture manifest: the 67 worked transactions of the Ledger Posting Design

One row per fixture, generated from the Ledger Posting Design itself (fixtures-manifest-v6.0.csv). Fixture = the test id; Tx = the worked transaction; Pattern = the section of the Ledger Posting Design; Source = LEDGER_TRANSACTION.source_type; Lines = posting lines the fixture asserts (0 for a worked case that posts nothing); Accounts = every account the fixture touches, in posting order; after every fixture the twelve identities F-ID1 to F-ID12 are checked; Page = where the expected postings are printed in the Ledger Posting Design v6.0; Status = build status at this edition. Patterns G (inter-group offsets, described by the entitlement view) and P (interruptions, C-INT1 to C-INT3) have no worked transaction of their own.

Fixture	Tx	Pat.	Source	Lines	Accounts	Page	Status
F-T1	T1	A	Collection	2	CL.ALI.USDT.COLLECTED CL.PROJECT.USDT.PAYABLE	8	planned (M2)
F-T2	T2	A	Conversion	6	CL.PROJECT.USDT.PAYABLE CL.ALI.USDT.COLLECTED CL.ALI.EUR.DUE CO.ALI.EUR.POOL CL.PROJECT.EUR.PAYABLE CO.NONE.EUR.EARN_GROSS	8	planned (M2)
F-T3a	T3a	A	Collection	2	CL.ALI.EUR.COLLECTED CL.PROJECT.EUR.PAYABLE	9	planned (M2)
F-T3	T3	A	Conversion	6	CL.PROJECT.EUR.PAYABLE CL.ALI.EUR.COLLECTED CL.ALI.EUR.DUE CO.ALI.EUR.POOL CO.NONE.EUR.EARN_GROSS	9	planned (M2)
F-T4	T4	B	Collection	2	CL.ALI.USDT.COLLECTED CL.PROJECT.USDT.PAYABLE	9	planned (M2)
F-T5	T5	B	Collection	2	CL.ALI.USDT.COLLECTED CL.PROJECT.USDT.PAYABLE	9	planned (M2)
F-T6	T6	B	Conversion	6	CL.PROJECT.USDT.PAYABLE CL.ALI.USDT.COLLECTED CL.ALI.EUR.DUE CO.ALI.EUR.POOL CL.PROJECT.EUR.PAYABLE CO.NONE.EUR.EARN_GROSS	10	planned (M2)
F-T7	T7	C	Settlement	2	CL.PROJECT.EUR.PAYABLE CL.ALI.EUR.DUE	10	planned (M2)
F-T8	T8	C	Confirmation	0	(no posting)	10	planned (M2)
F-T9	T9	C	Confirmation	2	CL.RECEIVER_X.EUR.SHORTFALL CL.PROJECT.EUR.PAYABLE	11	planned (M2)
F-T10	T10	C	BankFeeEvent	4	CO.NONE.EUR.EXP_BANKFEE CO.ALI.EUR.POOL CL.ALI.EUR.DUE CL.RECEIVER_X.EUR.SHORTFALL	11	planned (M2)
F-T11	T11	C	Conversion	8	CL.PROJECT.USDT.PAYABLE CL.ALI.USDT.COLLECTED CL.ALI.THB.DUE CO.ALI.THB.POOL CO.NONE.THB.EXP_ROUNDING CL.PROJECT.THB.PAYABLE CO.NONE.THB.EARN_GROSS	12	planned (M2)
F-T12	T12	D	Collection	2	CL.OWN.USDT.HELD CL.PROJECT.USDT.PAYABLE	12	planned (M2)
F-T13	T13	D	Reroute	4	CL.JETON.USDT.COLLECTED CL.OWN.USDT.HELD CO.OWN.USDT.WALLET CO.JETON.USDT.POOL	13	planned (M2)
F-T14	T14	D	Conversion	7	CL.PROJECT.USDT.PAYABLE CL.JETON.USDT.COLLECTED CL.JETON.EUR.DUE CO.JETON.EUR.POOL CO.NONE.EUR.EXP_PARTNER CL.PROJECT.EUR.PAYABLE CO.NONE.EUR.EARN_GROSS	13	planned (M2)
F-T15	T15	D	Collection	2	CL.OWN.USDT.HELD CL.PROJECT.USDT.PAYABLE	14	planned (M2)
F-T16	T16	D	Settlement	0	(no posting)	14	planned (M2)
F-T17	T17	D	Confirmation	4	CL.ALI.USDT.COLLECTED CL.OWN.USDT.HELD CO.NONE.USDT.EXP_NETWORK CO.OWN.USDT.WALLET	14	planned (M2)
F-T18	T18	E	Collection	2	CL.ALI.USDT.COLLECTED CL.PROJECT.USDT.PAYABLE	14	planned (M2)
F-T19	T19	E	Adjustment	2	CL.PROJECT.USDT.PAYABLE CL.ALI.USDT.COLLECTED	14	planned (M2)
F-T20	T20	E	Collection	2	CL.ALI.USDT.COLLECTED CL.PROJECT.USDT.PAYABLE	15	planned (M2)
F-T21	T21	E	Adjustment	3	CL.PROJECT.USDT.PAYABLE CL.ALI.USDT.COLLECTED	15	planned (M2)
F-T22a	T22a	F	Collection	2	CL.ALI.USDT.COLLECTED CL.PROJECT.USDT.PAYABLE	15	planned (M2)
F-T22	T22	F	Conversion	6	CL.PROJECT.USDT.PAYABLE CL.ALI.USDT.COLLECTED CL.ALI.GBP.DUE CO.ALI.GBP.POOL CL.PROJECT.GBP.PAYABLE CO.NONE.GBP.EARN_GROSS	15	planned (M2)
F-T23	T23	F	Settlement	2	CL.PROJECT.GBP.PAYABLE CL.ALI.GBP.DUE	16	planned (M2)

F-T24	T24	F	SettlementReturn	6	CL.ALI.GBP.DUE CL.PROJECT.GBP.PAYABLE CO.NONE.GBP.EXP_BANKFEE CO.ALI.GBP.POOL	16	planned (M2)
F-T24a	T24a	F	Conversion	4	CO.NONE.USDT.FX_RESULT CO.ALI.USDT.POOL CO.ALI.GBP.POOL CO.NONE.GBP.FX_RESULT	16	planned (M2)
F-T25	T25	F	Conversion	4	CL.PROJECT.GBP.PAYABLE CL.ALI.GBP.DUE CL.ALI.USDT.DUE CL.PROJECT.USDT.PAYABLE	17	planned (M2)
F-T26	T26	H	Conversion	2	CL.ALI.SGD.DUE CL.PROJECT.SGD.PAYABLE	17	planned (M2)
F-T27	T27	H	BalanceConversion	4	CL.PROJECT.SGD.PAYABLE CL.ALI.SGD.DUE CL.ALI.USDT.DUE CL.PROJECT.USDT.PAYABLE	17	planned (M2)
F-T28	T28	I	Collection	3	CL.ALI.USDT.COLLECTED CL.PROJECT.USDT.PAYABLE CL.SENDER_A.USDT.CREDIT	18	planned (M2)
F-T29	T29	I	Collection	3	CL.ALI.USDT.COLLECTED CL.SENDER_A.USDT.CREDIT CL.PROJECT.USDT.PAYABLE	18	planned (M2)
F-T30	T30	J	MarkupShare	2	CO.NONE.EUR.EARN_GROSS CO.PARTY_P.EUR.SHARE_PAYABLE	19	planned (M2)
F-T31	T31	J	MarkupShare	2	CO.PARTY_P.EUR.SHARE_PAYABLE CO.OWN.EUR.WALLET	19	planned (M2)
F-T32	T32	J	Conversion	8	CL.PROJECT.USDT.PAYABLE CL.AQN.USDT.COLLECTED CL.AQN.EUR.DUE CO.RAEEN.EUR.SHARE_RECEIVABLE CO.NONE.EUR.EARN_GROSS CO.NONE.EUR.EXP_PARTNER CL.PROJECT.EUR.PAYABLE	19	planned (M2)
F-T33	T33	J	Rebate	2	CO.AQN.USDT.REBATE_RECEIVABLE CO.NONE.USDT.EARN_REBATE	20	planned (M2)
F-T34	T34	J	Rebate	3	CO.OWN.USDT.WALLET CO.NONE.USDT.EARN_REBATE CO.AQN.USDT.REBATE_RECEIVABLE	20	planned (M2)
F-T35	T35	J	MarkupShare	2	CO.OWN.EUR.WALLET CO.RAEEN.EUR.SHARE_RECEIVABLE	20	planned (M2)
F-T36	T36	K	Conversion	7	CL.PROJECT.USDT.PAYABLE CL.ALI.USDT.COLLECTED CL.ALI.EUR.DUE CO.ALI.EUR.POOL CO.NONE.EUR.VAR_CONVERSION CL.PROJECT.EUR.PAYABLE CO.NONE.EUR.EARN_GROSS	21	planned (M2)
F-T37	T37	K	Conversion	7	CL.PROJECT.USDT.PAYABLE CL.ALI.USDT.COLLECTED CL.ALI.EUR.DUE CO.ALI.EUR.POOL CO.NONE.EUR.VAR_CUTOFF CL.PROJECT.EUR.PAYABLE CO.NONE.EUR.EARN_GROSS	21	planned (M2)
F-T38	T38	K	Conversion	7	CL.PROJECT.USDT.PAYABLE CL.ALI.USDT.COLLECTED CL.ALI.EUR.DUE CO.ALI.EUR.POOL CO.NONE.EUR.VAR_RATE_HONOUR CL.PROJECT.EUR.PAYABLE CO.NONE.EUR.EARN_GROSS	22	planned (M2)
F-T39	T39	L	Collection	2	CL.ALI.USDT.COLLECTED CL.PROJECT.USDT.PAYABLE	22	planned (M2)
F-T40	T40	L	LossEvent	5	CO.NONE.USDT.EXP_LOSS CO.ALI.USDT.POOL CO.OWN.USDT.WALLET CL.ALI.USDT.COLLECTED	23	planned (M2)
F-T41a	T41a	L	LossRecovery	2	CO.ALI.USDT.LOSS_RECEIVABLE CO.NONE.USDT.EXP_LOSS	23	planned (M2)
F-T41b	T41b	L	LossRecovery	2	CO.SENDER_A.USDT.LOSS_RECEIVABLE CO.NONE.USDT.EXP_LOSS	23	planned (M2)
F-T41c	T41c	L	LossRecovery	4	CO.ALI.USDT.POOL CO.NONE.USDT.EXP_LOSS CL.PROJECT.USDT.PAYABLE CL.ALI.USDT.COLLECTED	24	planned (M2)
F-T42	T42	L	LossRecovery	2	CO.ALI.USDT.POOL CO.ALI.USDT.LOSS_RECEIVABLE	24	planned (M2)
F-T43	T43	L	LossRecovery	2	CO.OWN.USDT.WALLET CO.SENDER_A.USDT.LOSS_RECEIVABLE	24	planned (M2)
F-T44	T44	M	Invoice	2	CO.NONE.EUR.RESELL_COST CO.VENDOR_V.EUR.VENDOR_PAYABLE	25	planned (M2)
F-T45	T45	M	Invoice	2	CO.CUSTOMER_C.EUR.RESALE_RECEIVABLE CO.NONE.EUR.RESELL_REVENUE	25	planned (M2)
F-T46	T46	M	Collection	4	CL.ALI.THB.COLLECTED CL.PROJECT.THB.PAYABLE CO.PROJECT.EUR.REMIT_CLAIM CO.CUSTOMER_C.EUR.RESALE_RECEIVABLE	26	planned (M2)

F-T47	T47	M	Conversion	6	CL.PROJECT.THB.PAYABLE CL.ALI.THB.COLLECTED CL.ALI.USDT.DUE CO.ALI.USDT.POOL CL.PROJECT.USDT.PAYABLE CO.NONE.USDT.EARN_GROSS	26	planned (M2)
F-T48	T48	M	Settlement	4	CL.PROJECT.USDT.PAYABLE CL.ALI.USDT.DUE CO.VENDOR_V.EUR.VENDOR_PAYABLE CO.PROJECT.EUR.REMIT_CLAIM	27	planned (M2)
F-T49	T49	M	Confirmation	2	CL.OWNDESK.USDT.DUE CL.ALI.USDT.DUE	27	planned (M2)
F-T50	T50	M	Settlement	8	CL.PROJECT.USDT.PAYABLE CL.OWNDESK.USDT.DUE CO.OWNDESK.USDT.POOL CO.NONE.USDT.VAR_FX_TIMING CO.VENDOR_V.EUR.VENDOR_PAYABLE CO.PROJECT.EUR.REMIT_CLAIM	28	planned (M2)
F-T51	T51	M	Settlement	11	CL.PROJECT.USDT.PAYABLE CL.OWNDESK.USDT.DUE CO.OWNDESK.USDT.POOL CO.NONE.USDT.VAR_FX_TIMING CO.OWN.USDT.WALLET CO.NONE.USDT.FX_RESULT CO.NONE.USDT.RESELL_REVENUE CO.NONE.EUR.FX_RESULT CO.PROJECT.EUR.REMIT_CLAIM	29	planned (M2)
F-T52a	T52a	N	Collection	2	CL.JETON.USDT.COLLECTED CL.PROJECT.USDT.PAYABLE	32	planned (M2)
F-T52	T52	N	Conversion	7	CL.PROJECT.USDT.PAYABLE CL.JETON.USDT.COLLECTED CL.JETON.EUR.DUE CO.JETON.EUR.POOL CO.NONE.EUR.EXP_PARTNER CL.PROJECT.EUR.PAYABLE CO.NONE.EUR.EARN_GROSS	32	planned (M2)
F-T53	T53	N	Collection	2	CL.ALI.USDT.COLLECTED CL.PROJECT.USDT.PAYABLE	32	planned (M2)
F-T54	T54	N	EarningsOffset	8	CL.JETON.EUR.DUE CL.PROJECT.EUR.PAYABLE CL.PROJECT.USDT.PAYABLE CL.ALI.USDT.COLLECTED CO.ALI.USDT.POOL CO.NONE.USDT.FX_RESULT CO.NONE.EUR.FX_RESULT CO.JETON.EUR.POOL	33	planned (M2)
F-T55	T55	N	Conversion	7	CL.PROJECT.USDT.PAYABLE CL.ALI.USDT.COLLECTED CO.ALI.USDT.POOL CO.NONE.USDT.EARN_GROSS CL.ALI.EUR.DUE CL.PROJECT.EUR.PAYABLE	34	planned (M2)
F-T56	T56	O	BalanceConversion	4	CL.PROJECT.EUR.PAYABLE CL.ALI.EUR.DUE CL.ALI.USDT.DUE CL.PROJECT.USDT.PAYABLE	35	planned (M2)
F-T57	T57	O	Reroute	4	CL.OWN.USDT.HELD CL.ALI.USDT.DUE CL.JETON.USDT.COLLECTED	35	planned (M2)
F-T58	T58	O	Conversion	7	CL.PROJECT.USDT.PAYABLE CL.JETON.USDT.COLLECTED CL.JETON.EUR.DUE CL.PROJECT.EUR.PAYABLE CO.NONE.EUR.EXP_PARTNER CO.NONE.EUR.VAR_CONVERSION CO.JETON.EUR.POOL	36	planned (M2)
F-T59	T59	Q	Adjustment	2	CO.NONE.EUR.EXP_BANKFEE CO.ALI.EUR.POOL	38	planned (M2)
F-T60	T60	R	Opening	2	CL.ALI.EUR.DUE CL.PROJECT.EUR.PAYABLE	39	planned (M2)
F-T61	T61	R	Opening	2	CO.ALI.EUR.POOL CO.NONE.EUR.OPENING_BALANCE	39	planned (M2)